

Project Demo Planning

Date	13 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Project Demo Planning

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement	Introduce the HDI concept and the need for a fast, interactive way to estimate it instead of the manual UNDP formula.	0.5	Nimmakayala Venkata Satheesh Kumar
2	Dataset & Architecture	Explain the UNDP HDI dataset (195 countries, 880 columns) and the dual-branch EnhancedHybridHDINet architecture (primary + secondary branches).	1.0	Cherukuri Sailaja Kiranmai
3	Model Training & Results	Discuss the AdamW/MSE training process and the results achieved: MAE 0.026, RMSE 0.031, 83% tier classification accuracy.	1.0	Nimmakayala Venkata Satheesh Kumar
4	Live Application Walkthrough	Screen-share the live application at hdi-webapp.onrender.com, select a country, adjust Life Expectancy/Schooling/GNI, and demonstrate the real-time HDI prediction.	1.5	Nimmakayala Venkata Satheesh Kumar
5	Deployment & Honest Limitations	Briefly cover the Render deployment setup and be upfront about the secondary-feature data-leakage limitation and the High-tier misclassification rate.	0.5	Cherukuri Sailaja Kiranmai
6	Conclusion & Future Scope	Wrap up and mention planned improvements, such as retraining the secondary branch on non-leaky indicators.	0.5	Cherukuri Sailaja Kiranmai

Demo Flow Summary

Step	Activity	Notes
1	Introduction & Problem Statement	Hook the audience by defining HDI and explaining why a hybrid neural network approach improves prediction speed over manual calculation.
2	Solution Overview	Briefly walk through the architecture diagram. Emphasize how decoupling the trained model (.pkl) from the Flask routing keeps the backend modular and scalable.
3	Live Feature Demonstration	Run two test cases live: a Very-High-HDI country (e.g., Norway) and a Low-HDI country (e.g., Afghanistan), then adjust GNI on one of them to show the what-if lever updating the score in real time.
4	Q&A Session	Ensure both team members are available to answer technical questions related to their assigned modules (Model/Training vs. Data/Deployment).